

Key concepts

Understanding platelet activation, the coagulation cascade, including amplification of thrombin, and the fibrinolytic process, is essential for understanding the pharmacology.

- Anticoagulants, fibrinolytics, and antiplatelet drugs act at specific points in the clotting cascade.
- The actions determine their therapeutic effects and adverse effects.
- The Notes Handout for this lecture includes detailed explanation of the cascades of reactions, defines the methods for monitoring activity, and the reversal agents. I also recommend watching a video on YouTube or other site.

Terms

aPTT	Activated partial thromboplastin time is for monitoring UFH and direct thrombin inhibitor therapy.
Anti-factor Xa assay	Chromogenic assay for monitoring the activity of factor Xa inhibitors
ACT	Activated clotting time, a rapid point-of-care test to monitor the activity of high-dose heparin and direct thrombin inhibitors
PT/INR	Prothrombin time is the most commonly used test for measuring warfarin therapy. International normalized ratio is a calculation for standardizing prothrombin time.
AT	Antithrombin

UFH	Unfractionated heparin
LMWH	Low molecular weigh heparin
HIT	Heparin-induced thrombocytopenia
DOAC	Direct oral anticoagulant
DTI	Direct thrombin inhibitor
DVT	Deep vein thrombosis
PE	Pulmonary embolism
VTE	Venous thromboembolism
STEMI	ST-elevation myocardial infarction
PCI	Percutaneous coronary intervention
tPA	Tissue plasminogen activator
PCC	Prothrombin complex concentrate (a human plasma product)

Antithrombotic drugs are used to prevent or treat blood clots. They are broadly categorized into three main types: **anticoagulants**, **antiplatelet drugs**, and **thrombolytics**.

Anticoagulants	Fibrinolytics	Antiplatelet Drugs
Warfarin	Thrombolytics	Antiplatelet Drugs
Heparin, unfractionated (UFH)	Alteplase	Aspirin
LMWHs	Reteplase	Clopidogrel
Enoxaparin	Tenecteplase (TNKase)	Prasugrel
Dalteparin	Streptokinase	Ticagrelor
Fondaparinux	Urokinase	Dipyridamole
Direct Thrombin Inhibitors		Cilostazol
Bivalirudin		
DOACs		
Factor Xa Inhibitors:		
Rivaroxaban (Xarelto),		
Apixaban (Eliquis),		
Edoxaban (Savaysa)		
Direct thrombin inhibitor		
Dabigatran (Pradaxa)		

Anticoagulants inhibit either the action of coagulation factors or prevent their synthesis.

- Heparin is an injectable, rapid-acting drug with an indirect anticoagulant action by accelerating the action of antithrombin (AT) by ~1000-fold.
 - Heparin has no intrinsic anticoagulation activity.
 - AT is a small glycoprotein that inhibits serine proteases of the contact activation pathway.
- Unfractionated heparin (UFH) is a natural product extracted from porcine intestinal mucosa. UFH is mixture of straight-chain, anionic glycosaminoglycans with a wide range of molecular weights.
- UFH reversibly binds with high affinity to a specific pentasaccharide sequence on AT, causing a conformational change that opens AT's catalytic site. Thrombin must also bind heparin at a site near the pentasaccharide site. AT inactivates thrombin, factor Xa, and other coagulation factors.

Anticoagulants: UFH

- Heparin prevents the progression of existing clots by preventing further clot formation. It has many clinical applications. UFH is administered as an IV loading dose followed by continuous infusion or subQ administration (prophylaxis use only). It binds many plasma proteins, neutralizing its activity and causing unpredictable pharmacokinetics. It is eliminated by the reticuloendothelial system (nonrenal elimination). UFH has a half-life of 1-2 hours after discontinuation. UFH therapy is monitored by aPTT, anti-factor Xa assay, and ACT.
- UFH may cause heparin induced thrombocytopenia – HIT. HIT type I is a mild reduction in platelets during the first two days of exposure, which spontaneously returns to normal and is not associated with thrombosis.
- HIT type II is an immune-mediated potentially life-threatening complication of heparin therapy when IgG attached to the heparin-PF4 complex binds the Fc receptor on the platelet surface, leading to platelet activation and a severe hypercoagulable state. Heparin must be discontinued and anticoagulation with a non-heparin anticoagulant initiated. HIT type II occurs usually 5 to 14 days after initiation of therapy.

- Low-molecular-weight forms of heparin (LMWHs), enoxaparin, dalteparin, and tinzaparin, complex with antithrombin at the pentasaccharide site, accelerating AT's inactivation of factor Xa. They have low antithrombin activity. Because of their shorter molecular sequence, they have low affinity for thrombin.
- LMWHs have several therapeutic uses including treatment and prevention of deep vein thrombosis (DVT), acute coronary syndromes, anticoagulation bridging, and adjunct to PCI. UFH and LMWHs are preferred anticoagulant therapy for pregnant patients.
- LMWHs are administered subcutaneously, have reliable absorption, renal excretion (dose adjustment for renal insufficiency), and biological half-life of ~4-6 hours. Onset of action is 20-30 min with full effect in about 4 hours. Monitoring is usually not required because of their predictable pharmacokinetics. If clinically appropriate, anti-factor Xa assay is used.
- There is a lower risk of HIT with LMWH use. Patients with HIT may have cross-sensitivity. LMWHs should not be used in patients with HIT.
- **Bleeding** is the chief adverse effect of the heparins – indeed, all anticoagulants and antiplatelet agents. Patients on anticoagulation or antiplatelet therapy are at increased risk of complications of **neuraxial anesthesia** techniques, esp. spinal epidural hematoma.
- Protamine sulfate is a specific antidote for reversal of heparin. It directly binds UFH, neutralizing its activity. LMWHs are less easily inactivated by protamine.

Anticoagulants: Fondaparinux

- Fondaparinux is a synthetic pentasaccharide that reversibly binds the pentasaccharide site on AT, leading to selective inactivation of factor Xa.
- Fondaparinux therapeutic uses include treatment and prophylaxis of DVT, PE, and prophylaxis in patients with HIT.
- It is administered subcutaneously, has reliable absorption, renal excretion (dose adjustment for renal insufficiency), and half-life of ~17 hours (1x daily administration).
- Monitoring is not required due to its predictable pharmacokinetics. If clinically appropriate, monitoring is by anti-factor Xa assay.
- Fondaparinux is not inactivated by protamine.

Direct-acting Oral Anticoagulants (DOACs): Factor Xa inhibitors

- The DOAC factor Xa inhibitors, rivaroxaban, apixaban, and edoxaban, bind selectively and reversibly to factor Xa, reducing production of thrombin from prothrombin.
- They are used for the treatment and prophylaxis of DVT and PE, post-surgical thromboprophylaxis, and non-valvular atrial fibrillation for prevention of stroke and systemic embolism. They should not be used for anticoagulation in patients with mechanical heart valves due to lack of evidence of safety and efficacy.
- Rivaroxaban and apixaban are metabolized by CYPs including CYP3A4. ***Strong CYP inhibitors and inducers should be avoided.*** Edoxaban is excreted mainly unchanged in the urine. All are substrates of P-glycoprotein. ***Strong P-glycoprotein inhibitors and inducers should be avoided.***
- Monitoring is generally not required. If clinically appropriate, anti-factor Xa assay is used.
- Bleeding is the most serious adverse effect. There is no specific antidote. PCC is used for reversal in patients with severe bleeding.

Direct-acting Oral Anticoagulants (DOACs): Direct thrombin inhibitor

- Dabigatran is an oral DTI included in the DOACs group. It inhibits both free and clot-bound thrombin by binding specifically and reversibly to the catalytic site of thrombin.
- Like the other DOACs, it is used for the treatment and prophylaxis of DVT and PE, VTE, and non-valvular atrial fibrillation, and may be used in patients with HIT. It is contraindicated in patients with mechanical heart valves and not recommended in those with bioprosthetic heart valves. Dabigatran demonstrated inferior efficacy and higher risk of thromboembolic events and bleeding compared to warfarin.
- Dabigatran is available as the prodrug dabigatran etexilate, which is hydrolyzed by plasma esterases to active dabigatran and excreted in urine. It is a P-glycoprotein substrate and ***P-glycoprotein strong inducers and inhibitors should be avoided.*** Therapeutic monitoring is not required.
- Idarucizumab is a dabigatran-specific reversal agent that neutralizes its anticoagulant action.

Parenteral Direct Thrombin Inhibitors

- Argatroban is a synthetic intravenous anticoagulant derived from L-arginine that reversibly binds thrombin's catalytic site. Its effects are immediate. It is used for the prophylaxis and treatment of thromboembolism in patients with HIT, and during PCI in patients who have or are at risk for developing HIT. Argatroban is metabolized in the liver and has a half-life of <1 hour.
- Bivalirudin is an analog of hirudin, a naturally occurring peptide in leech salivary glands. Bivalirudin has a bifunctional action; it specifically and reversibly binds to thrombin's catalytic site and exosite 1. It is used for anticoagulation in patients with HIT, undergoing PCI, and cardiac surgery as an alternative to heparin. Bivalirudin is metabolized by proteolysis and has a half-life of about 25 minutes (normal renal function).
- Anticoagulation monitoring is done using aPTT or ACT.
- PCC is used for reversal of severe bleeding.

Anticoagulants: Vitamin K Antagonists

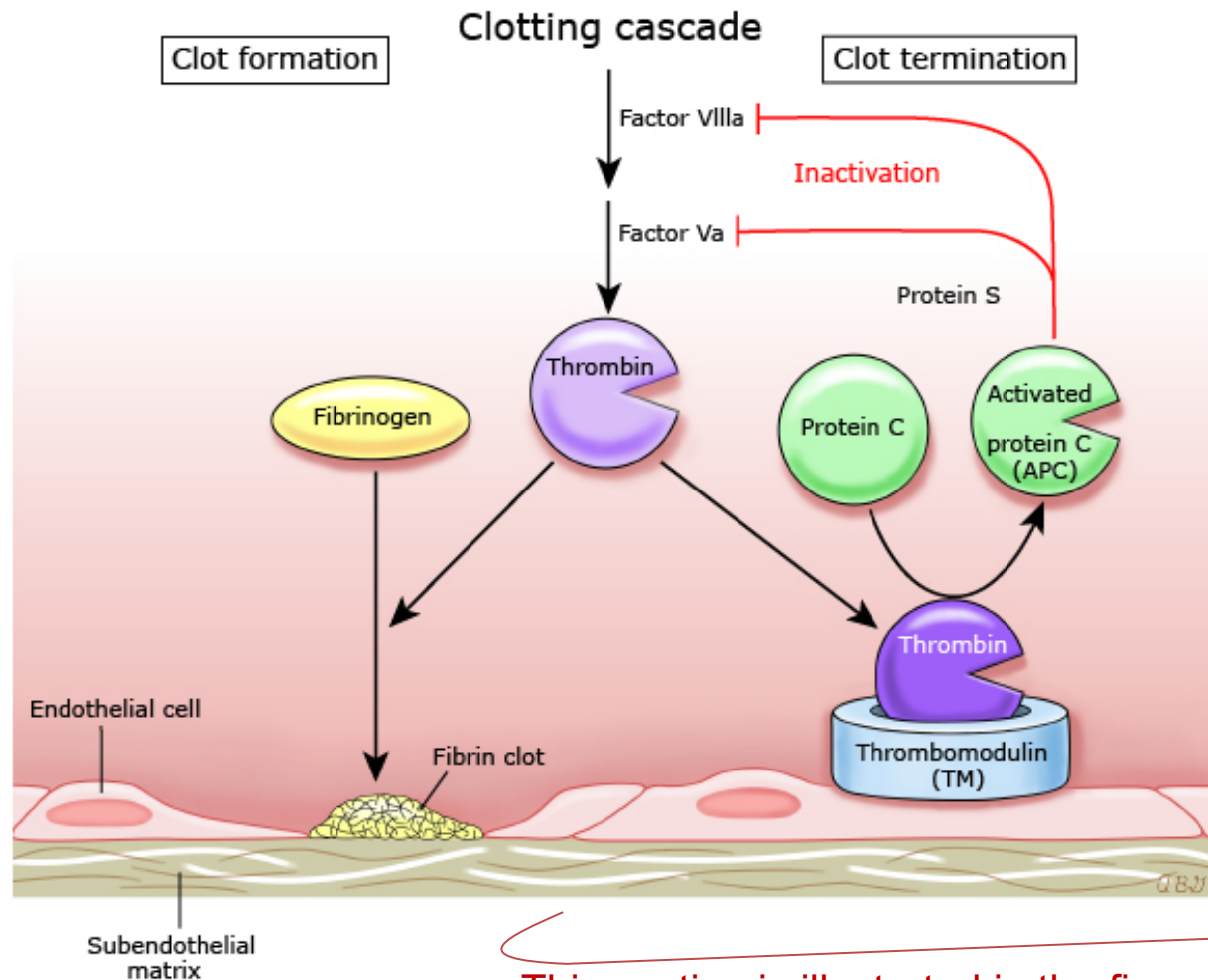
Warfarin is the only coumarin anticoagulant available in the US.

- Factors II (prothrombin), VII, IX, and X, which are synthesized by the liver, require vitamin K as a cofactor in the glutamation of a number of their residues by gamma-carboxyglutamase. The gamma-carboxyglutamyl residues bind calcium ions, which are essential for interaction between the coagulation factors and platelet membranes.
- In the carboxylation reactions, the reduced vitamin K cofactor is converted to vitamin K epoxide during the reaction. Vitamin K is regenerated from the epoxide by vitamin K epoxide reductase (VKOR1C), the enzyme that is inhibited by warfarin, which reduces the activity of these clotting factors by up to 40%. Anticoagulant effects are delayed for 72 to 96 hours – the time required to deplete the pool of circulation clotting factors.
- Therefore, bridging therapy with a rapid acting anticoagulant (UFH, LMWH, or DOAC) may be required.

Warfarin

- Warfarin has many therapeutic uses, such as prevention of thromboembolism in patients with atrial fibrillation with or without prosthetic heart valves, treatment and prophylaxis of DVT and PE, and others.
- Warfarin is available as a racemic mixture taken orally with rapid complete absorption. It is highly bound to plasma proteins. Warfarin crosses the placenta. The active S-isomer is metabolized mainly by CYP2C9 to inactive metabolites, which are excreted in urine and feces. Numerous drug-food and drug-drug interactions may increase or decrease warfarin's effect.
- Bleeding is the major adverse effect. Vitamin K administration orally or IV is used to reverse bleeding. Replacement of coagulation factors with PCC or fresh frozen plasma may be need for severe bleeding.
- Rare complications are skin lesions and necrosis and cholesterol emboli causing purple (blue) toe syndrome.
- Warfarin is teratogenic and contraindicated in pregnancy.

Hemostatic control mechanisms and termination of clotting



This section is illustrated in the figure.

Refer to UpToDate for an overview of hemostasis regulation and the roles of specific procoagulant and anticoagulant factors in clinical thrombosis and hemostasis.

APC: activated protein C; TM: thrombomodulin.

Courtesy of Lawrence LK Leung, MD

UpToDate®

NATURAL ANTICOAGULANT MECHANISMS

Regulators of blood coagulation

- **Tissue factor pathway inhibitor (TFPI):** A plasma protein that binds to the TF-VIIa after activation of factor X to Xa.
- **Antithrombin (AT):** A small plasma glycoprotein synthesized in the liver and circulating in plasma that inactivates several enzymes of the coagulation system, particularly thrombin and factor Xa (also factors IXa, XIa, XIIa).

Pharmacology relevance: Heparin sulfate drugs, like endogenous glycosaminoglycans, amplify AT action ~1000-fold.

- **Thrombomodulin:** A high affinity thrombin receptor on the endothelial cell membrane that forms a complex with thrombin and removes thrombin from circulation.
- **Protein C:** A vitamin K-dependent plasma zymogen requires activation by thrombomodulin-bound thrombin. Activated protein C is a protease that, with cofactor protein S, inactivates the FVa and FVIIIa. It also inhibits plasminogen activator inhibitor-1 (PAI-1) and PAI-2, thereby facilitating fibrinolysis.
- **Protein S:** Vitamin K-dependent glycoprotein; Cofactor of protein C
- **Nitric oxide (NO) and prostacyclin (PGI₂):** Synthesized by endothelial cells and released into the blood induce vasodilation and inhibit platelet activation and subsequent aggregation.

Heparin (UFH) Pharmacokinetics:

A	I.V. infusion: onset within minutes subQ: erratic absorption (only for prophylaxis) onset 20-30 minutes	
D	– Nonspecific binding endothelial cells and plasma proteins – Polarity and large size prohibits crossing of cell membranes	– Does not cross blood-brain barrier – Does not cross the placenta – Does not enter breast milk
E	Nonrenal elimination – Eliminated by the reticuloendothelial system (mainly liver and spleen)	
t _{1/2}	Mean 1.5 h (adults) after completion of infusion Half-life is affected by hepatic cirrhosis, obesity, malignancy, presence of pulmonary embolism, and infections	

Monitoring UFH anticoagulation: aPTT, anti-factor Xa assay, ACT

Because of protein and tissue binding, free heparin levels are variable after administration

Monitor platelet counts due to potential for heparin-induced thrombocytopenia.

Heparin Goal of Therapy

To reduce blood coagulability to a level that is low enough to prevent thrombosis but not so low as to promote spontaneous bleeding

Heparin prevents progression of existing clots by inhibiting further clotting.

UFH Therapeutic Uses (many)	Adverse Effects
Rapid onset of action • DVT and PE treatment and prevention • Acute MI or unstable angina, initial mgmt. • Cardiopulmonary bypass surgery • Renal dialysis • Pregnancy: UHF, LMWH, or fondaparinux are the preferred agents	• Bleeding – the major adverse effect of anticoagulants and antiplatelet agents. • Neuraxial anesthesia techniques → spinal or epidural hematomas → pressure on spinal cord or cauda equina → paralysis • Heparin-induced thrombocytopenia (HIT) – Type I: Benign, 1-2 days of therapy, spontaneous return to normal – Type II: Immune mediated; life-threatening • Osteoporosis (long-term use)

Heparin resistance: Very high heparin dose is needed to achieve therapeutic aPTT or ACT

- Congenital or acquired AT deficiency
- Elevated levels of plasma proteins that bind heparin
- Elevated acute phase proteins, factor VIII and fibrinogen

LMWH / Fondaparinux PK, Therapeutic uses, and Cautions

LMWHs SubQ; reliable absorption; renal excretion; biological t $\frac{1}{2}$ 4-5 hours	Fondaparinux SubQ; reliable absorption; renal excretion; biological t $\frac{1}{2}$ ~17 hours
Use: • DVT / PE treatment and prophylaxis • Acute coronary syndromes • Warfarin bridging • PCI adjunctive use • Cautions: as for unfractionated heparin	Uses: • DVT / PE treatment and prophylaxis • HIT anticoagulation • Acute thrombosis unrelated to HIT in patients with a history of HIT • Acute symptomatic superficial vein thrombosis (≥ 5 cm in length) of the legs
Routine anticoagulation monitoring usually not required. Anti-factor Xa-assay if clinically indicated.	
Adverse Effects • Immune-mediated HIT occurs less frequently with LMWHs • Cross-sensitivity may occur • Avoid in patients with HIT	• Adverse effects • Avoid fondaparinux alone in PCI due to guided catheter thrombosis. The fondaparinux molecule is too short to block the thrombotic process.

Common: Bleeding; Avoid in neuraxial anesthesia.

Pharmacokinetics of Direct Factor Xa Inhibitors

	Rivaroxaban	Apixaban	Edoxaban
A	Oral	Oral	Oral
D	Protein binding >90% P-glycoprotein substrate	Protein binding ~90% P-glycoprotein substrate	Protein binding ~50% P-glycoprotein substrate
M	Hepatic CYP3A4/5 Bioavailability 66-100% dose-dependent	CYP3A4/5 predominantly Bioavailability 50%	Minimal metabolism Bioavailability ~60%
E	Renal, ~35% unchanged and fecal excretion	Renal, ~30 % unchanged and fecal excretion	Urine unchanged
t _{1/2}	5-9 hours 11-13 hours in elderly 2x daily treatment, 1x daily prophylaxis	~8 hours (lower dose) ~15 hours (higher dose) 2x daily dose for most indications	10-14 hours once daily dosing
Avoid use or adjust dose in patients with impaired renal function			

Direct-acting Factor Xa Inhibitors

Mechanism of action	Direct, selective and reversible inhibition of free and clot-bound factor Xa, independent of AT.
Therapeutic uses	DVT and PE treatment and prophylaxis Post-operative (knee, hip) thromboprophylaxis Atrial fibrillation, nonvalvular, prevention of stroke and systemic embolus
Routine anticoagulation monitoring usually not required. Anti-factor Xa if clinically indicated.	
Adverse effects	Bleeding Neuraxial anesthesia hematoma risk
Reversal Update: Andexanet-alfa was recently withdrawn from U.S. market. Risks of thromboembolic events outweigh the benefits.	Recombinant inactivated FXa: Andexanet-alfa binds and sequesters FXa inhibitors rivaroxaban and apixaban ➤ Caution: Andexanet-alfa also inhibits tissue factor pathway inhibitor, increasing tissue factor-initiated thrombin generation. Serious and life-threatening arterial and venous thrombotic, ischemic, and cardiac events have been observed 3 to 30 days after use.
Drug interactions	Other drugs that affect coagulation and platelet function CYP3A4 and P-glycoprotein inhibitors and inducers

Dabigatran etexilate: Oral Direct Thrombin Inhibitor

PK	Etexilate oral prodrug rapidly converted to active dabigatran Moderate protein binding; P-glycoprotein substrate Rapid complete hydrolysis of etexilate plasma and hepatic esterases → dabigatran hepatic glucuronidation → active acylglucuronide isomers Bioavailability 3-7% Renal excretion of dabigatran acylglucouronides t _{1/2} ~16 hours	
MOA	Reversible selective inhibition of catalytic site of free and clot-bound thrombin	
Uses	DVT and PE prevention and treatment Atrial fibrillation, nonvalvular for prevention of thrombotic complications Contraindicated in patients with mechanical heart valves; not recommended in patients with bioprothetic valves – reduced efficacy, excess incidence of major bleeds	
Monitoring	Routine monitoring not required. Test with aPTT if clinically indicated.	
Reversal	Idarucizumab binds specifically to dabigatran and glucuronide metabolites.	
Cautions	Bleeding Neuraxial anesthesia hematoma risk Other drugs affecting coagulation Pgp inducers: contraindicated Pgp inhibitors: monitor therapy	Pregnancy (limited data) Patients ≥75 year – extreme caution Obesity (BMI≥40 kg/m ²): Use alternative drug

Pharmacology of Argatroban and Bivalirudin

Argatroban

synthetic, derived from L-arginine

IV, immediate onset, moderate protein binding, minor hepatic metabolism (CYP3A4/5), feces unchanged and metabs., $t_{1/2}$ 40-50 min

Uses:

Anticoagulation patients with HIT
PCI

Bivalirudin

synthetic, 20 amino acid peptide

IV, immediate onset, binds only thrombin, proteolytic cleavage, urine (20%), $t_{1/2}$ ~25 min

Uses:

Anticoagulation patients with HIT
PCI
Cardiac surgery alternative to heparin

Anticoagulation monitoring: aPTT, ACT

Bleeding, neuraxial anesthesia hematoma risk, acute stent thrombosis within 4 hours in patients with STEMI undergoing PCI; hypersensitivity reactions reported with hirudin derivatives

Desirudin: subQ for DVT prophylaxis following hip surgery | Not available for clinical use

Warfarin Pharmacokinetics

A	Oral, rapid complete absorption; I.V. 🔑 Dose according to PT/INR results	Onset of anticoagulation: 24-72 h Full therapeutic effect 5-7 days Duration of action 2-5 days
D	>99% bound to albumin	🔔 Crosses placenta
M	Hepatic: S-warfarin : mainly CYP2C9 <i>R-warfarin</i> predominantly via CYP3A4 Inactive metabolites	
E	Urine, metabolites t _{1/2} 20-60 hours (highly variable)	
Monitoring: PT: prothrombin time and INR: international normalized ratio		
High potential for drug-drug and drug-food (Vit K containing foods) interactions		

Factor VII	Factor IX	Factor X	Factor II	Protein C	Protein S
$t_{1/2}$ 6 h	$t_{1/2}$ 24 h	$t_{1/2}$ 36 h	$t_{1/2}$ 50 h	$t_{1/2}$ 8 h	$t_{1/2}$ 30 h

Because of the long $t_{1/2}$ of some of the coagulation factors, in particular factor II (thrombin), ***the full anti-thrombotic effect of warfarin is not achieved for several days.*** PT may be prolonged soon after administration due to the more rapid reduction of factors with a shorter $t_{1/2}$.

Pharmacogenomics: Variable responses

Variability in warfarin response: *CYP2C9* and *VKORC1* genetic variants

CYP2C9 reduced function alleles → decreased rate of metabolism → increased risk of bleeding in patients who carry at least one copy of a reduced function allele

VKORC1 variations can significantly alter warfarin pharmacodynamics leading to warfarin resistance or warfarin sensitivity 1639A or 1173T allele → lower warfarin dose

🔑 When determining the initial dose, other patient related factors must also be taken into account; for example: body weight, concomitant medications, and comorbidities.

Warfarin Therapeutic Uses

Many anticoagulation uses:

- Thromboembolic disorders
such as, venous thromboembolism and pulmonary embolism
- Prosthetic heart valves
- Atrial fibrillation
- Adjunct to reduce risk of systemic embolism after myocardial infarction
such as, recurrent MI, stroke
- Acute venous thromboembolism

Bridging with UFH, LMWH, DOAC

Warfarin is not useful for rapid anticoagulation in emergency situations.

- PT / INR measurement is most sensitive to deficiencies in the vitamin K-dependent clotting factors
- Prothrombin time (PT):
 - Measures time for fibrin clot formation
 - Evaluates extrinsic and common coagulation pathways
- International normalized ratio (INR):
 - a calculation for standardizing prothrombin time
 - $INR = [Patient\ PT / Normal\ mean\ PT]^{ISI}$
 - Range for patients on warfarin: 2.0 to 3.5 (depending on indication for use)

Warfarin: Adverse Effects

Effect	Mechanism
Bleeding	Extension of warfarin's mechanism of action • Risk of intracranial hemorrhage increases dramatically with an INR >4 • especially in older patients
Thrombosis and skin necrosis	Low level of protein C or protein S • Starting warfarin therapy in patients with acute thrombosis without concomitant heparin or other immediately acting anticoagulant • Inherited deficiency of protein C or protein S
Purple toe syndrome	Cholesterol microembolization • Warfarin therapy may release atheromatous plaque emboli

Warfarin resistance:

Patients requiring >20 mg/day:

- Excessive vitamin K intake
- *VKORC1* variant alleles leading to warfarin resistance

Apparent "resistance"

- Noncompliance / Laboratory error

Warfarin Sensitivity

Patients requiring <1.5 mg/day:

- Reduced function alleles
- *CYP2C9* or *VKORC1* haplotypes
- Lower maintenance dose required
- Increase risk of bleeding

Warfarin CONTRAINDICATIONS for Use

- Active bleeding
- Hemorrhagic tendency
- Heparin induced thrombocytopenia
- Severe uncontrolled or malignant hypertension
- Bacterial endocarditis

These patients are at risk of bleeding complications

- Dissecting aortic aneurysm
- Pericarditis or pericardial effusion
- Lumbar block anesthesia
- Pregnancy
- Hypersensitivity to warfarin or any component of the formulation

Drug Interactions High potential! Warfarin is subject to more clinically significant drug interactions than any other drug in use today

- Anticoagulant effect may be:
 - **Reduced**, permitting thrombosis
 - **Increased**, causing hemorrhage
- **The most serious interactions are those that increase the anticoagulant effect**

Summary of properties of the oral anticoagulants

FEATURES	WARFARIN	RIVAROXABAN	DABIGATRAN
Bioavailability (%)	100%	80-100%	3-7% (etexilate)
Metabolism	CYP2C9, 1A2, 3A4, 2C19	CYP3A4/5, 2J2 P-glycoprotein substrate	Hepatic glucuronidation P-glycoprotein substrate
t _{1/2} (h)	20-60h (highly variable)	5-9h	12-17h
Renal excretion	Yes (mainly metabolites)	Yes, including unchanged Dose adjustment in renal impairment	Yes, largely unchanged Dose adjustment in renal impairment
Antidote effect	Vitamin K	Andexanet-alfa; Not dialyzable	Idarucizumab; ~60% dialyzable
Therapeutic uses	- DVT and PE, prophylaxis and acute treatment - Coronary ischemia/acute MI - Atrial fibrillation, patients with chronic a-fib or prosthetic heart valves	- DVT/PE, prophylaxis/treatment - Non-valvular atrial fibrillation: prophylaxis of stroke and systemic embolism	- DVT / PE prophylaxis/treatment - Non-valvular atrial fibrillation: prophylaxis of stroke and systemic embolism
Drug Interactions	High potential	Drugs that affect platelet function or coagulation CYP3A4, P-glycoprotein related interactions	Drugs that affect platelet function or coagulation P-glycoprotein inhibitors